

KAPREKAR CONTEST – FINAL – SUB-JUNIOR

AMTI – OCTOBER, 2005.

Instructions:

1. Answer as many questions as possible.
2. Elegant and novel solutions will get extra credits.
3. Diagrams and explanations should be given wherever necessary.
4. Attach the filled-in FACE SLIP and your rough working sheets along with the answer script.
5. Maximum time allowed is THREE hours.

1. Let N be the greatest integral multiple of 8, such that no two of its digits are the same. What is the remainder when N is divided by 1000?
2. Show that it is impossible to find a positive integer such that the sum of its square and its cube is an integral multiple of the square of the next higher integer.

3. If $a \neq 0, b \neq 0, c \neq 0$ and if

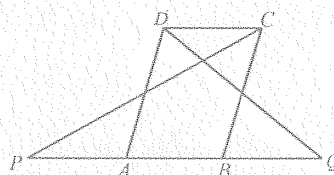
$$\frac{1}{a} + \frac{1}{b} + \frac{1}{a+x} = 0$$

$$\frac{1}{a} + \frac{1}{c} + \frac{1}{a+y} = 0$$

$$\frac{1}{a} + \frac{1}{x} + \frac{1}{y} = 0$$

find the value of $a+b+c$.

4. The side AB of the parallelogram $ABCD$ is produced both ways to P and Q such that $PA=AB=BQ$. Given that $AD=2AB$, find the angle between DQ and CP .



5. Find all positive integers ' n ' such that each of the numbers $n+1, n+3, n+7, n+9, n+13$ and $n+15$ is a prime.
6. In how many ways two squares can be selected from a 8×8 chessboard so that they are not in the same row or same column?
7. (a) Find the possible remainders when the cube of any integer is divided by seven.
(b) Show that $abc(a^3 - b^3)(b^3 - c^3)(c^3 - a^3)$ is divisible by 7.
8. Between the digits 4 and 9 several fours and after them the same number of eights were inserted. For example between 4 and 9 we may insert three fours followed by three eights 444888 to get 44448889. Prove that the resulting number is always a perfect square.

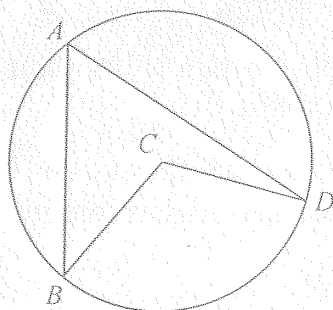
9. From the set $\{1, 2, 3, \dots, 2005\}$ in how many ways can one choose two different numbers whose sum is an even number?

10. Can the numbers $\frac{14x+5}{9}$ and $\frac{17x-5}{12}$ both be integers for some integer x ?

11. Find all integers such that $2x^2 - x - 36$ is the square of a prime number.

12. When I open my mathematics book, there are two pages that face me. If the product of the two page numbers is 2756, what are the two page numbers?

13.



A walker's path in a park is as shown in the diagram. One can walk all along the circumference or along the line segments marked in the figure. A child walks along the circumference-path starting at A and after going round returns to A . Her grandfather starts at A and walks through the route $AB \rightarrow BC \rightarrow CD \rightarrow DA$ and thus returns to A . Who walked more distance, the child or the grandpa?

14. The number $A837B$ is divisible by 88. Find the digits A and B .

15. Suppose for integers x, y we have $2x+3y$ divisible by 17; prove that $9x+5y$ is also divisible by 17.

16. (a) Can the sum of 2005 odd numbers be zero?

(b) Let $a_1, a_2, a_3, \dots, a_n$ be any rearrangement of $1, 2, 3, \dots, n$. (For example 2, 4, 3, 1, 6, 5 is a rearrangement of 1, 2, 3, 4, 5, 6). If n is odd prove that $(a_1 - 1)(a_2 - 2)(a_3 - 3) \dots (a_n - n)$ is an even number.

17. Find all possible values of n such that a rectangular board $9 \times n$ can be covered with tiles of the shape



18. A triangle ABC has its circumcentre at O and M is the midpoint of the median through A . If OM is produced to N such that $OM = MN$, prove that N lies on the altitude through A .